

An Active-Toe-Coupled Spring–Tendon Energy Recirculation System for Bipedal Locomotion

Wenyu Li, Zhiyuan Xu, Yuelong Zhang, Jiaying Li, Jingang Yi, and Tao Liu

Abstract—This paper presents a Spring–Tendon Energy Recirculation System (STERS) that couples the ankle and an active toe through a shared elastic pathway in bipedal locomotion. STERS enables cross-joint energy storage, transfer, and release throughout multi-stage foot–ground contact. A multi-joint parallel elastic actuator is designed to generate the tendon-induced ankle and toe torques. An ankle-centered stiffness optimization and reinforcement learning methods are further used for the dominant energy-recirculation pathway. Hardware ablation experiments on two distinct bipedal platforms, BioLeg and Roboto Origin, show that the complete STERS configuration reduces leg energy consumption by 44.3% and 36.3%, respectively, compared with their single-segment flat-foot baselines. STERS also reduces the peak driving torque at the ankle by approximately 75%, while improving impact recovery, heading preservation, and the smoothness of foot–ground contact transitions. These results demonstrate the effectiveness and cross-platform transferability of STERS to improving both energy efficiency and stability in bipedal locomotion.

Index Terms—Parallel elastic actuator, bipedal robot, gait, spring tendon, reinforcement learning.

I. INTRODUCTION

Bipedal robots have demonstrated remarkable locomotion capabilities in traversing diverse terrains while maintaining robust balance and achieving a human-like gait [1]–[4]. Bio-inspired design has shown significant potential in key robotic areas, including joint arrangement [5], actuation architecture [6], and elastic element implementation [7]. Existing studies have largely investigated passive elasticity and active toe actuation as separate design directions [8], [9]. In contrast, most active-toe robots treat the toe as an independent distal degree of freedom for improving support stability, push-off capability, or motion versatility [10]–[13].

In mammalian hindlimbs, the Achilles tendon and plantar fascia (AT-PF) do not function as isolated elastic tissues but form a continuous force-transmission pathway through the calcaneus and foot structures [14] [15] [16]. This anatomical continuity suggests that an active robotic toe should not be treated solely as an independent distal degree of freedom, but should be incorporated into a unified elastic pathway spanning

the ankle and toe joints. Applying these biological mechanisms to bipedal robot platforms remains nontrivial, as exploiting the toe joint and tendon-coupled structure is often accompanied by a sacrifice in controllability and walking stability [7] [17] [18] [19]. Therefore, a key unresolved challenge is how to exploit these biological advantages while preserving controllable and stable walking.

To address this knowledge gap, we propose a Spring–Tendon Energy Recirculation System (STERS) that adopts an active-toe-coupled biarticular design. The ankle serves as the primary energy-input and load-regulation joint, while the toe regulates the pathway geometry, contact transition, and late-stance energy release. In the STERS design, the coupled elastic behavior is formulated as a multi-joint parallel elastic actuator (PEA) system and embedded into the reinforcement learning (RL) environment as a physics-informed prior. This integration enables the policy to exploit the spring-tendon dynamics during multi-contact-stage locomotion.

The main contributions of this work are summarized as follows: (1) Unlike previous toe-joint designs that treat the toe as an isolated distal degree of freedom (DOF), the new STERS uses the active toe as part of a coupled energy storage-and-release mechanism; (2) Using a multi-joint PEA model, we analyze and derive the tendon-induced ankle and toe torques from the ankle-toe coupling geometry. Based on this model, we further develop an ankle-centered stiffness optimization method that targets the dominant energy-recirculation pathway during spring loading; (3) We develop a physics-aware reinforcement learning framework by explicitly applying the model-estimated tendon torques within the simulation environment and introducing contact-stage gait guidance for the segmented foot; and (4) we validate the complete system on both the BioLeg and Roboto Origin platform through hardware ablation experiments. The results demonstrate reduced walking energy consumption and peak ankle torque, together with improved trajectory tracking and impact recovery.

The rest of the paper is organized as follows. We discuss the related work in Section II. We present the overall design of the STERS in Section III. Modeling and control design of the STERS are presented in Sections IV and V, respectively. Experiments and results are presented in Section VI. Finally, we conclude the paper in Section VII.

II. RELATED WORK

A. Energy Recirculation Systems in Legged Robots

Substantial effort has been devoted to reducing energy loss during locomotion using different methods. BirdBot [7]

This work was supported in part by the NSFC under Grant 52175033 and Grant U21A20120 (*Corresponding author: Tao Liu*)

W. Li, J. Li and T. Liu are with the State Key Laboratory of Fluid Power and Mechatronic Systems, School of Mechanical Engineering, Zhejiang University, 310027, Hangzhou, China (e-mail: wenyuli@zju.edu.cn; 12225069@zju.edu.cn; liutao@zju.edu.cn).

Z. Xu and Y. Zhang are with the ZJU-UIUC Institute, International Campus, Zhejiang University, 314400, Haining, China (e-mail: zhiyuan2.24@intl.zju.edu.cn; yuelong.24@intl.zju.edu.cn).

J. Yi is with the Department of Mechanical and Aerospace Engineering, Rutgers University, Piscataway, NJ 08854 USA (e-mail: jgyi@rutgers.edu).

and SPEAR [18] are representative examples that imitate biological tendon function through spring-cable systems and achieve efficient locomotion via gait-phase energy recirculation. Related ideas have also been explored in other tendon-inspired robotic systems and wearable devices [8], [17], [20]. Beyond these bio-inspired designs, complex energy-saving mechanisms, including PEAs with decoupling clutches, switchable variable-stiffness actuators, and hydrostatic load-carrying assistance, have also been developed to improve the efficiency of legged robots [19], [21]–[23].

Although these studies have demonstrated promising reductions in the energy cost of legged locomotion, most of them are subject to a critical limitation: the integration of energy recirculation systems often introduces additional kinematic constraints and increased inertia, thereby substantially complicating the control of bipedal robots. Under conventional model-based control frameworks, these increased systems complexity often prevents locomotion algorithms from efficient deployment and computational feasibility. Consequently, most existing studies remain limited to single-leg demonstrations or sagittal-plane-constrained walking, and have not yet demonstrated fully autonomous bipedal locomotion. In this work, we adopt an RL-based control strategy by embedding the energy recirculation model into the learning environment, enabling the agent to learn and adapt to the dynamics of the energy recirculation mechanism and ultimately achieve stable, independent bipedal locomotion.

B. Modeling of Parallel Elastic Actuators

Compared with more complex energy recirculation mechanisms, PEA systems are generally lighter and easier to integrate, and have therefore been widely adopted in legged robots as compliant actuation or energy-saving mechanisms. Accordingly, accurate modeling of PEA structures is crucial for both model and RL-based control strategies. Mazumdar et al. [9] implemented carefully designed constant-stiffness parallel springs at the hip and ankle joints of the STEPPR biped. By incorporating spring torque into the control framework and with the model-based feedforward compensation, STEPPR achieved unsupported three-dimensional (3D) treadmill walking. In [8], the spring-tendon mechanism was modeled as a prismatic joint, with a spring constant corresponding to the actual robot spring. Filip et al. [24] introduced a novel method for optimizing the design parameters of a PEA-driven leg. In the optimization framework, the teacher policy directly observed the design parameters during training, enabling the policy to adapt to different spring configurations.

In a recent example, the E-Go robot [25] incorporated PEA models by analyzing their dynamic stiffness and compensating for the resulting forces within the whole-body control framework. Ding et al. [26] also showed that explicit modeling of parallel elasticity, together with hybrid torque control, can improve both performance and energy efficiency in dynamic jumping. In this work, the spring-tendon mechanism of STERS extends from the calf, wraps around the heel, and inserts at the toe, thereby constituting a multi-joint PEA. To facilitate control and learning, we represent this mechanism as a sim-

plified multi-input-multi-output (MIMO) system that is readily integrated into the RL environment.

C. Active Toe Joint Design for Bipedal Locomotion

The functional role of the toe joint has long been recognized in biological locomotion, especially in the transition from heel strike to push-off and in maintaining stable support during stance [14]–[16]. Inspired by these observations, active toe joints have been incorporated into robotic designs to enhance lower-limb motion capability. Nishiwaki et al. [11] designed toe joints for robots to achieve knee-down motion, walking speed augmentation, and high-step climbing. Kouchak and Mokhtarian [12] analyzed the controllability of a bipedal robot with active toe joints and proved that the robot can maintain stability in the sagittal plane using only the ankle and toe joints. Liu et al. [13] examined human walking with a specific focus on toe motion, while Tlalolini et al. [10] further investigated the function of the toe during human walking and proposed a method to achieve an optimal walking gait by optimizing the ground reaction force (GRF) and joint torque. Collectively, these studies demonstrate that active toe joints play an important role in bipedal locomotion performance. However, these analyses of toe joint function remain largely intuitive and lack comprehensive quantitative metrics for evaluating the performance of the system.

Existing studies have demonstrated that toe joints improve the kinematic capability, support stability, and push-off behavior of bipedal robots. However, most prior works treat the toe joint as an isolated additional degree of freedom, rather than as an integrated component of a tendon-coupled elastic energy-recirculation pathway. As a result, the interaction among active toe actuation, multi-joint elastic force transmission, and contact-stage coordination during bipedal walking remains insufficiently explored. In this work, the active toe joint is not merely added to increase foot dexterity; instead, it is mechanically coupled with the ankle through STERS and coordinated by a reinforcement learning policy guided by a multi-contact-stage gait. This formulation enables the robot to exploit toe-assisted push-off and tendon-mediated energy recirculation in a unified locomotion framework.

III. OVERALL DESIGN OF THE STERS

A. Mechanism Design

Fig. 1(a) shows the mammalian hind-limb structure that inspires the proposed force-transmission mechanism, with the highlighted region indicating the tendon-based energy pathway. Fig. 1(b) illustrates the detailed STERS structure. STERS comprises two core components: an active toe joint and a spring-tendon mechanism. The active toe joint provides an additional DOF at the distal end of a conventional planar foot. It is driven by a motor mounted near the ankle via a timing belt. The second core component, the passive spring-tendon mechanism, utilizes a steel cable running from the calf, past the heel, to the toe joint, thereby functioning as a passive artificial AT-PF structure. Its path is guided by four V-groove bearings, with the foot sole itself acting as a distal sheath. The proximal end of the cable is connected to the compression

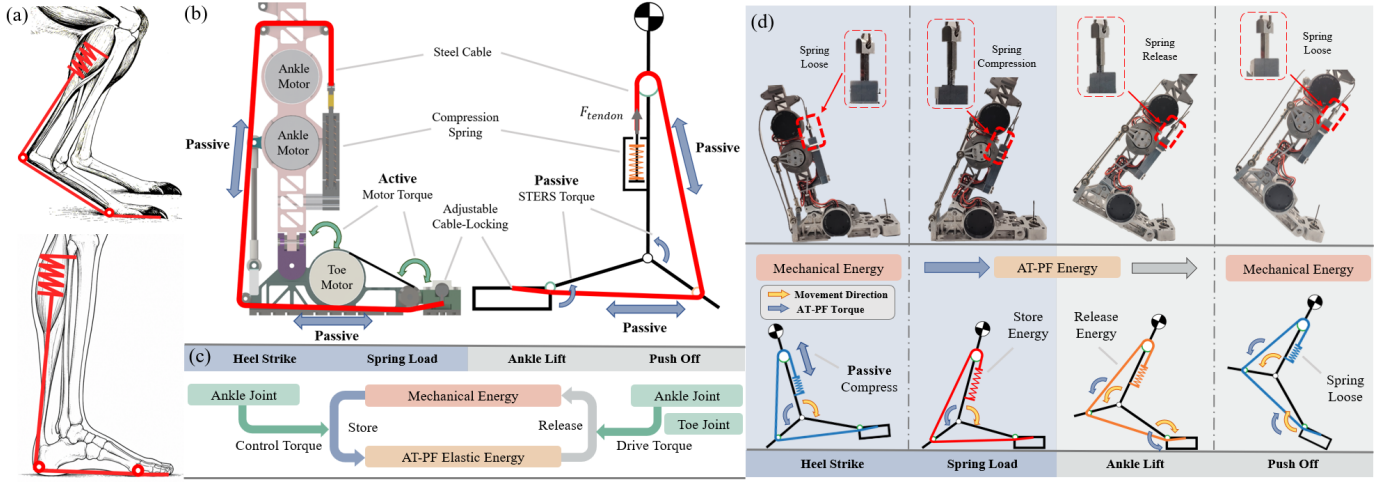


Fig. 1. The design of STERS and the process of multi-contact-stage gait. (a) Elastic AT-PF architecture of human and kangaroos as biological inspiration. (b) The mechanical structure of STERS. (c) The energy flow in the multi-contact-stage gait. (d) The process of multi-contact-stage gait and the corresponding state of AT-PF spring element.

spring unit mounted on the calf, while its distal end is fixed to an adjustable cable-locking device on the toe segment. **The ankle module is actuated by two DM-J4340P-2EC brushless joint motors (DAMIAO, China), each equipped with a 40:1 gearbox and providing a rated torque of 9 N·m, a peak torque of 27 N·m, and a rated power of approximately 33.9 W. The two motors form a two-DOF parallel actuation mechanism, whose coupled motions generate the ankle roll and pitch rotations.** Overall, the STERS can be readily integrated into many existing planar-foot bipedal robots, requiring only minor modifications to their original mechanical structure.

B. Energy Recirculation Pathway

To clarify how the AT-PF structure and active toe contribute to locomotion, we interpret the conventional heel-to-toe stance process using four contact-related stages: heel strike, spring load, ankle lift, and push off. Fig. 1(c) illustrates the energy flow of the multi-segment foot with the STERS and Fig. 1(d) shows the AT-PF spring elements in these stages. A stage-based interpretation of the energy-recirculation pathway and ankle-toe coordination are enabled within a conventional heel-to-toe walking pattern.

In the heel strike and spring load stages, since the whole foot is in contact with the ground, the toe joint remains nearly stationary. As the robot COM moves forward, ankle dorsiflexion stretches the tendon pathway and elastic element. Therefore, the spring-tendon mechanism absorbs mechanical work from the ankle motion and stores it as elastic energy. Meanwhile, the ankle torque generated by the STERS partially offsets the COM-induced gravitational torque, thereby reducing the support torque required from the ankle actuators (muscle or motors) and allowing them to mainly provide regulation torque.

In the ankle lift stage, the ankle joint extends and its motion direction becomes consistent with the torque caused by STERS. As a result, the stored elastic energy is released through the ankle joint, reducing the actuator effort required to

lift the heel and push the foot away from the ground. During this stage, the STERS torque at the toe joint mainly enhances contact stability by increasing the toe-ground interaction. Notably, the toe-side moment arm of the AT-PF structure is substantially smaller than the ankle-side moment arm [14]. Therefore, the overall energy flow of robot is dominated by the ankle-side interaction, even though the STERS performs local negative work at the toe joint. Subsequently, during the Push Off stage, the ankle approaches its extended configuration and the toe joint actively pushes against the ground. Hence, the remaining elastic energy stored in the STERS spring is released through the coordinated ankle-toe motion, providing additional assistive torque to the actuators until the stored energy is fully released.

In the next section, we present the modeling and analysis of the STERS design.

IV. MODELING AND ANALYSIS OF THE STERS

A. STERS Force/Torque Model

To maintain balance and exploit the energy-recirculation capability, the STERS control policy must account for the passive elastic torques generated by the spring-tendon mechanism. Fig. 2(a) shows the schematic of the STERS design. The tendon path starts from the constant-stiffness spring mounted on the calf, passes around the heel pulley, and attaches to the toe segment, thereby forming a multi-joint PEA. Unlike conventional single-joint PEAs, STERS spans both the ankle and toe joints. Therefore, ankle rotation and toe rotation jointly change the total tendon length and determine the tendon force generated by the shared spring. This tendon force is then mapped to the ankle and toe joints through their corresponding effective moment arms, producing the ankle and toe torques, denoted as τ_a^s and τ_t^s , respectively.

We denote the ankle and toe angles as θ_a and θ_t , respectively. We consider θ_a and θ_t as inputs and τ_a^s and τ_t^s as outputs. As shown in Fig. 2(a), we denote l_1 and l_2 as the lengths of the calf and heel links, respectively, and θ_0 is the initial angle between the calf and the heel. STERS

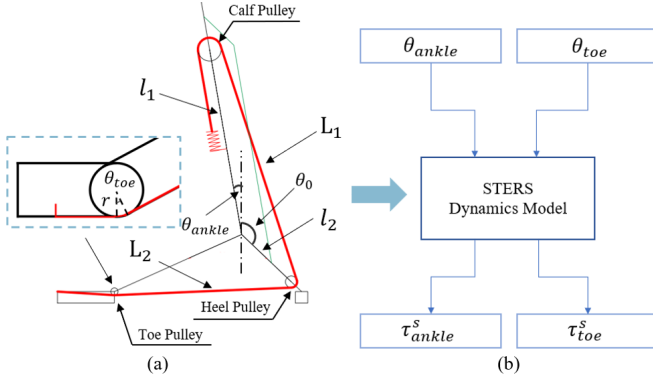


Fig. 2. Spring-Tendon Energy Recirculation System. (a) Schematic diagram of STERS. (b) MIMO representation of the STERS dynamics model

is designed to operate only when $\theta_a > 0$, because the spring-cable system provides tension force only. Furthermore, we denote L_1 and L_2 as the tendon lengths from the calf pulley to the heel pulley, and from the heel pulley to the toe pulley, respectively. Then, L_1 in the initial and current configurations are given by $L_1 = \sqrt{l_1^2 + l_2^2 - 2l_1l_2 \cos(\theta_0)}$ and $L_1' = \sqrt{l_1^2 + l_2^2 - 2l_1l_2 \cos(\theta_0 + \theta_{ankle})}$, respectively. The variation of L_1 is then $\Delta L_1 = L_1' - L_1$. Since L_2 changes only with toe rotation, its variation is given by $\Delta L_2 = r_{toe}\theta_t$, where r_{toe} represents the radius of the toe pulley.

The elastic tendon force is computed as $F_t = k(\Delta L_1 + \Delta L_2)$, where k is the stiffness coefficient of the spring.

Based on the tendon-routing geometry, the effective moment arms at the ankle and toe joints are calculated as

$$r_a = \frac{l_1 l_2 \sin(\theta_0 + \theta_a)}{L_1'}, \quad r_t = r_{toe}, \quad (1)$$

The corresponding tendon-induced torque magnitudes are then given by

$$\tau_a^s = r_a F_t, \quad \tau_t^s = r_t F_t. \quad (2)$$

In addition, different STERS behaviors can be obtained by tuning the spring stiffness k . Increasing k enhances the energy-storage capacity of the spring-tendon mechanism, but also increases its resistance to ankle and toe dorsiflexion. The stiffness should be selected to balance energy recirculation and the additional resistance introduced to joint motion.

B. Spring Stiffness Optimization

During the Spring Load stage of the multi-contact-stage gait, STERS provides an additional tendon-induced ankle torque τ_a^s in (??). This torque acts together with the ankle motor to counteract the robot COM-induced gravitational torque, denoted as τ_{COM} . By selecting an appropriate spring stiffness for STERS, the motor supporting work during this process can be minimized, allowing the ankle motor to mainly provide control torque rather than continuously resisting the gravitational torque. Therefore, the objective of the spring stiffness optimization is to reduce the residual supporting work required from the ankle motor

Since STERS stores energy mainly during the Spring Load stage, we select a single straight leg with the knee

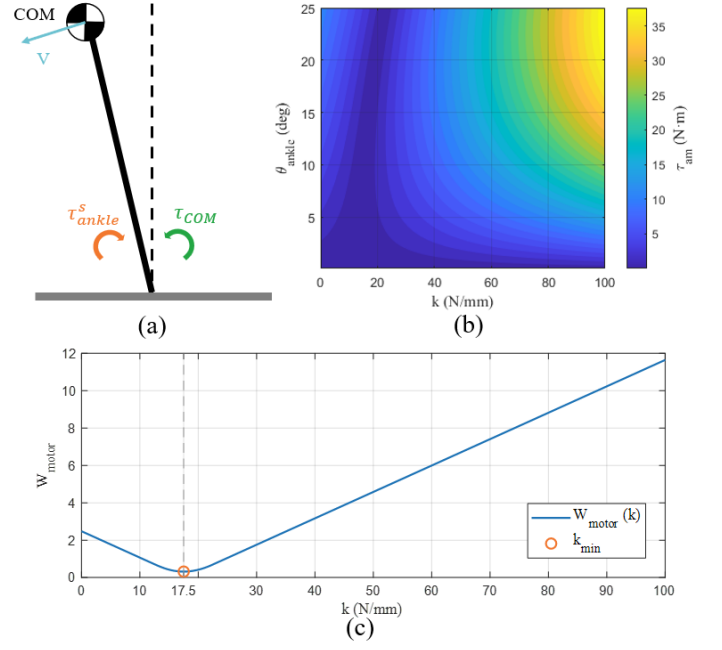


Fig. 3. Optimization process of spring stiffness in STERS (a) Inverted pendulum model for a single leg. (b) Torque map of τ_{am} . There exists a valley along the θ_{ankle} axis. (c) The local minimum of W_{motor} and the optimized k^* .

fully extended as the analysis subject, such that the hip and knee motions are neglected in the stiffness optimization. During this stage, the whole foot remains in contact with the ground, and the toe joint is assumed to remain stationary. Joint friction and air resistance are neglected. Furthermore, for the purpose of stiffness optimization, we approximate the stance motion as quasi-steady walking, where the robot maintains an approximately constant forward COM speed and the ankle angular acceleration during stance is neglected.

Subsequently, we model this stance leg as an inverted pendulum pivoting at the ankle pitch joint, along with the virtual torsion spring from STERS at the ankle, as shown in Fig. 3(a). Then, we can list the terms of Lagrangian:

$$L = \frac{1}{2} m L_C^2 \dot{\theta}_a^2 - mg L_C \cos \theta_a + \frac{1}{2} k \Delta L_1^2 \quad (3)$$

where L_C represents the length of the pendulum. Furthermore, the Lagrange's equations can be derived:

$$\tau_a^m = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_a} \right) - \frac{\partial L}{\partial \theta_a} \quad (4)$$

And the torque of the ankle motor τ_a^m can be obtained:

$$\tau_a^m = M + G, \quad (5)$$

where $M = m L_C^2 \ddot{\theta}_a$, and $G = mg L_C \sin \theta_a - k \frac{\Delta L_1}{L_1'} (l_1 l_2 \sin(\theta_0 + \theta_a)) = \tau_C - \tau_a^s$. The first term in G corresponds to the gravitation-induced torque $\tau_C = mg L_C \sin \theta_a$ and the second term corresponds to the additional tendon-induced ankle torque τ_a^s in (??). Therefore, G represents the residual torque after partial compensation by the STERS:

$$G = \tau_C - \tau_a^s \quad (6)$$

Under the constant-speed walking assumption, the ankle angular acceleration during stance phase is neglected and therefore, we approximate the inertial torque $M = 0$.

Furthermore, integrating $|\tau_a^m|$ with respect to θ_a , we obtain the positive work provided by the ankle motor during the stance phase as

$$W_{motor} = \int_0^{\theta_{SL}} |\tau_a^m| d\theta_a = \int_0^{\theta_{SL}} |\tau_C - \tau_a^s| d\theta_a.$$

Since STERS generates tendon force only when the tendon is tensioned, its stiffness-dependent contribution is considered within the ankle dorsiflexion range of the Spring Load stage, denoted as $[0, \theta_{SL}]$. In this range, the COM-induced gravitational torque and the STERS torque act in opposite directions at the ankle pitch joint, and the motor provides residual control torque after this compensation.

As a result, the positive work performed by the ankle motor, denoted as W_{motor} can be treated as a function of the spring stiffness k , and the optimized stiffness is obtained by minimizing $W_{motor}(k)$:

$$k^* = \arg \min_{k \in [0, 200]} W_{motor}(k). \quad (7)$$

The range $k \in [0, 200]$ is selected sufficiently large to ensure that the minimum of W_{motor} is covered within the optimization range.

V. PHYSICS-AWARE RL CONTROL

A. Learning Framework

STERS introduces two coupled challenges for locomotion control: configuration-dependent ankle-toe elastic torques and segmented foot-ground contact transitions. Although the proposed multi-contact-stage gait defines the desired coordination pattern of STERS, it does not directly provide a feedback control law for the coupled ankle-toe system. The active toe joint introduces segmented foot-ground contact, while the spring-tendon mechanism generates configuration-dependent torques at both the ankle and toe joints. As a result, the locomotion dynamics depend strongly on contact transitions and tendon-induced joint torques, making purely model-based control difficult to design.

Reinforcement learning is adopted to handle the hybrid and coupled dynamics introduced by STERS. Unlike a purely model-based controller, the RL policy can learn coordinated locomotion behaviors through interaction with a simulation environment that explicitly includes segmented foot-ground contact and tendon-induced ankle-toe torques. We formulate the locomotion task as a continuous-control Markov Decision Process, where the policy observes the robot state and outputs joint commands to maximize the expected cumulative reward. In this work, the learning environment is designed around two key components: incorporation of the analytical STERS dynamics and multi-contact-stage gait guidance, which are described in the following subsections.

The policy is trained in the NVIDIA Isaac Lab environment. It achieves a simulation throughput of approximately 55,000 steps per second on an NVIDIA GeForce RTX 4060 GPU, running 4,096 parallel environments. The overall training framework is illustrated in Fig. 4.

B. Physics-Based Incorporation of STERS Dynamics

STERS includes a spring-tendon mechanism that introduces configuration-dependent passive torques at both the ankle and toe joints. This mechanism cannot be properly represented by reward shaping alone, because its effect is not a high-level behavioral preference but a physical interaction directly imposed on the robot dynamics. Therefore, we incorporate the analytical STERS model into the simulation environment as a physics-informed prior.

As shown in Fig. 4, the incorporation process consists of five steps. First, θ_a and θ_t are obtained from the simulation environment. Second, these joint angles are used to compute the geometric change in tendon length. The tendon force F_t is calculated from the optimized spring stiffness k^* , that is, $\text{clip}(0, k^*(\Delta L_1 + \Delta L_2))$. Finally, the resulting STERS torques τ_a^s and τ_t^s are applied to the corresponding joints as additional efforts:

$$\tau_{joint} = \tau_p + \tau_a \quad (8)$$

where τ_p and τ_a represent the passive and active robot joint torques, respectively. And τ_{joint} is the final torque of matrix robot joints. Additionally, τ_p is equal to τ_i^s only at the ankle and toe joint positions, and is zero elsewhere. This formulation separates active motor control and passive elastic assistance. The policy still outputs joint position commands through the PPO controller, while the STERS model provides additional passive torques to the ankle and toe joints inside the physics simulation. As a result, the policy learns locomotion under the explicit influence of the spring-tendon mechanism, rather than treating this mechanism as an unmodeled disturbance. To improve robustness against modeling errors and hardware variation, domain randomization is further applied to the spring stiffness, the initial ankle angle, and other physical parameters during training.

C. Contact-Stage Guidance for the Segmented Foot

The gait guidance terms are introduced because the reward framework used for the single-segment planar foot cannot be directly transferred to the two-segment foot with an active toe joint. We propose two key reward terms that guide the agent to walk consistently with the multi-contact-stage gait: step reward and gait-consistency reward.

The step reward is designed to encourage the robot to take steps by evaluating how and when each part of the foot contacts the ground, and it is the fundamental reward term that decides whether the agent can walk. Since the toe joint mechanically divides the foot into two distinct segments, we let the superscript indicate the foot segment (e.g., t for toe, h for heel) and the subscript describe the contact state (e.g., a for air, c for contact). For example, t_a^t represents the current duration that the toe has been in the air. For a single foot, we define the contact detection time vector as $T = [t_a^{\max} \ t_c^{\max} \ t_a^{\min} \ t_c^{\min}]$, with

$$t_i^{\max} = \max(t_i^t, t_i^h), \quad t_i^{\min} = \min(t_i^t, t_i^h), \quad i = a, c$$

During a gait cycle, each two-segment foot can be classified into three contact states: whole foot contact (WFC), whole foot

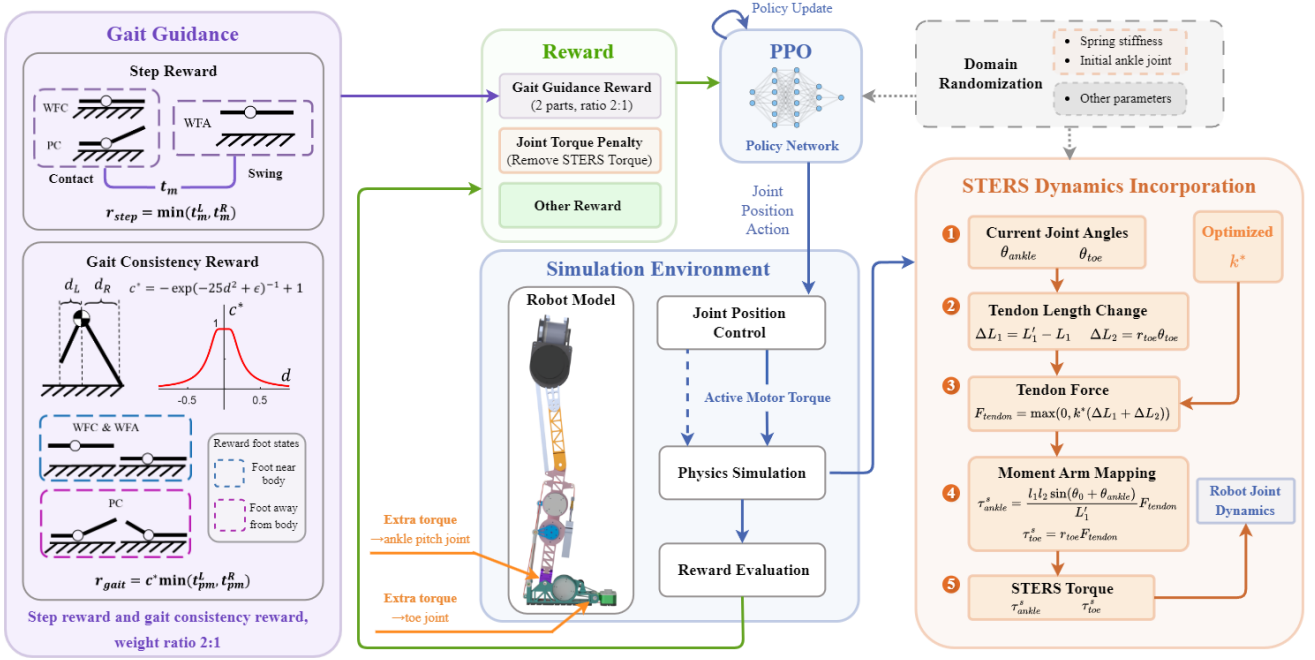


Fig. 4. Physics-aware reinforcement learning framework. The analytical STERS model is incorporated as physics-informed prior. The virtual spring torque provided by STERS is obtained after domain randomization on k and θ , and is then directly applied to the corresponding joints of the simulation model. The gait guidance part is composed of the step reward and gait reward terms at a weight ratio of 2:1.

in the air (WFA), and partial contact (PC). To represent these states compactly, we define the sign vector of the contact-time vector $\mathbf{T}$ as ${}_s\mathbf{T} = \text{sign}(\mathbf{T})$. Each element of ${}_s\mathbf{T}$ indicates whether the corresponding contact or air duration is nonzero. The mapping between the three contact states and the resulting sign-vector patterns is summarized in (9).

$${}_s\mathbf{T} = \begin{bmatrix} {}_s\mathbf{T}_{WFC} \\ {}_s\mathbf{T}_{WFA} \\ {}_s\mathbf{T}_{PC} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \quad (9)$$

where the subscript of ${}_s\mathbf{T}$ describes the three different contact states. Therefore, in the step reward term, we determine whether a single foot is in contact with the ground by checking whether $t_c^{\max} > 0$. Additionally, the current contact time and air time are denoted as $t_c^{\max}$ and $t_a^{\min}$ respectively. Furthermore, the function of step reward term is defined as:

$$r_{\text{step}} = \min(t_m^R, t_m^L) \quad (10)$$

where R and L represent the right and left legs, respectively, and t_m is denoted as:

$$t_m = \begin{cases} t_c^{\max}, & (t_c^{\max} > 0, t_a^{\min} = 0) \\ t_a^{\min}, & (t_a^{\min} > 0) \\ 0, & \text{else} \end{cases} \quad (11)$$

Thanks to the step reward, the agent can walk step by step. However, it does so using a gait that is implicitly learned by the agent itself. Thus, we add a gait-consistency reward that aims to encourage the agent to walk consistently with the multi-contact-stage gait proposed in Section III.B.

As described previously, the multi-contact-stage gait divides the single stance phase into four stages. By analyzing these stages, we conclude that the most effective feature for guiding

the agent to learn the designed gait is the change in contact state. As illustrated in Fig. 5(a), during the stance phase of that gait, the foot contact state follows a PC-WFC-PC sequence. Therefore, in the gait-consistency reward term, this sequence is represented as a bell-shaped curve with a flat top in the reward assigned to the WFC and WFA state. Specifically, within the gait-consistency reward function, this trend is described by a variation factor of the horizontal distance from the ankle joint to the body in the sagittal plane:

$$c^* = -\exp\left(-\frac{1}{25d^2 + \epsilon}\right) + 1 \quad (12)$$

where d represents the distance between the ankle and hip of the support leg in the sagittal plane, and ϵ is an extremely small number to avoid infinite values. The relationship between c^* and contact state is illustrated in Fig. 5(b). The term c^* is introduced to encourage full-foot ground contact when the foot is positioned directly beneath the body during stance, while leaving the contact pattern unconstrained during heel strike and toe push-off as the foot moves away from the body.

The complete function of the gait-consistency reward is

$$r_{\text{gait}} = c^* \min(t_{pm}^R, t_{pm}^L) \quad (13)$$

where the subscript pm represents a single stance phase with the support foot entirely in contact with the ground, therefore t_{pm} is denoted as:

$$t_{pm} = \begin{cases} t_c^{\min}, & (t_c^{\min} > 0) \\ t_a^{\min}, & (t_a^{\min} > 0) \\ 0, & \text{else} \end{cases} \quad (14)$$

Another reason for adding the gait-consistency reward is to compensate for the defects introduced by the step reward.

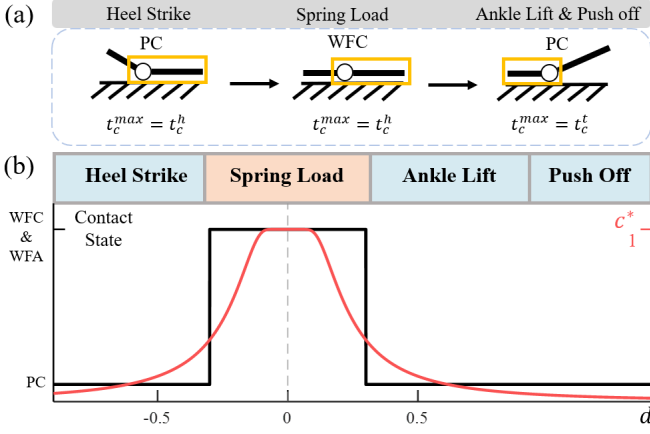


Fig. 5. Contact state during stance phase in multi-contact-stage gait. (a) Transition of the max contact time. The yellow box represents the location of $t_c^{\max}$. (b) The relationship between c^* and contact state.

Fig. 5(a) illustrates the theoretical contact process of the multi-contact-stage gait during the stance phase. In the first two stages, $s_{t_c}^{\max}$ always represents the contact time of the heel. However, during the transition into the Ankle Lift stage, the heel lifts, and t_c^h drops to zero. Consequently, $s_{t_c}^{\max}$ undergoes a sudden decline as it shifts to t_c^t . The gait-consistency reward term addresses this by assigning additional reward to this contact process, thereby compensating for the decline. Thus, we summarize the roles of the step reward and the gait-consistency reward as basic and compensatory: the step reward encourages stepping behavior, whereas the gait-consistency reward further guides the robot toward the designed multi-contact-stage gait. The weight ratio of the step reward to the gait-consistency reward is 2:1.

VI. EXPERIMENTS AND RESULTS

We designed and conducted experiments on two distinct bipedal platforms: BioLeg and Roboto Origin. The impact experiment was designed to examine static stability, disturbance recovery, and the stabilizing contribution of the active toe joint and spring-tendon mechanism. The walking experiment was designed to evaluate energy recirculation, gait characteristics and tracking performance during locomotion. Together, these two experiments allow us to isolate the effects of the active toe joint and the spring-tendon mechanism on both stationary robustness and dynamic walking performance.

A. Experimental Setup and Ablation Configuration

The BioLeg is a bipedal robot developed by our team. It possesses a total of 12 DOF. Each leg is equipped with 6 DOFs: hip roll and pitch, knee flexion, ankle roll and pitch, and an active toe joint. The control policy of BioLeg is built upon the reward framework used in [27], with some modifications to adapt it to our system. The dimensions of BioLeg are scaled to approximate those of a human child, and the structure is primarily constructed of aluminum alloy, resulting in a total weight of 16.21 kg.

The Roboto Origin is an open-source humanoid robot developed by RoboParty, China. It has six degrees of freedom

TABLE I
EXPERIMENT CONFIGURATION

Configuration	Without spring-tendon mechanism	With spring-tendon mechanism
Without active toe joint	Group 1	-
With active toe joint	Group 2	Group 3

in each leg, including hip roll, yaw, and pitch, knee flexion, and ankle roll and pitch. To evaluate the transferability of STERS and the associated physics-aware reinforcement learning method, we replaced Roboto's original single-segment planar foot with a multi-segment foot featuring an active toe joint. We also incorporated a gait-guidance term and the STERS dynamics model into Roboto's original reinforcement learning framework.

It should be noted that Roboto is equipped with a more mature and comprehensive reinforcement learning framework and is therefore expected to exhibit better overall performance than BioLeg. However, Roboto was introduced not to compare the absolute performance of the two robotic platforms, but to evaluate the transferability of the STERS mechanical structure and its associated learning method across different robots. To avoid confounding effects arising from differences in robot hardware, baseline control frameworks, and other platform-specific factors, the results in this experimental section are compared only among different configurations on the same robotic platform. No direct cross-platform performance comparison is made between BioLeg and Roboto.

To isolate the individual and combined effects of the active toe joint and spring-tendon mechanism, we designed an ablation study with three configurations for both BioLeg and Roboto, as listed in Table I.

- Group 1 (baseline): In this group, the robots equipped with a single-segment planar foot and its corresponding training strategy were used as the experimental baseline.
- Group 2: The robots are equipped with an active toe joint. Their policies are trained with the gait guidance terms. This guidance is introduced because both baseline reward frameworks used for the single-segment planar foot did not produce stable simulated walking after the active toe joint was added.
- Group 3: The robot is equipped with complete STERS, including an active toe joint and the spring-tendon mechanism. The RL environment of this group builds upon Group 2 by only integrating the STERS dynamics model. Additionally, the training process introduces randomization to the spring stiffness k and the initial ankle pitch angle θ_0 .

A configuration with the spring-tendon mechanism but without the active toe joint was not included, because the current STERS structure uses the toe segment as the distal tendon attachment and therefore requires the active toe module to complete the tendon path.

Additionally, we also tested the robot with complete STERS hardware using the same policy in Group 2. In this case, without injecting the virtual STERS torque into the train-

ing process, the robot was able to maintain static standing but failed to generate stable stepping motion (see video). This qualitative result, as shown in the supplementary video, indicates that explicitly incorporating STERS dynamics is necessary for learning a walking policy on the complete spring-tendon system.

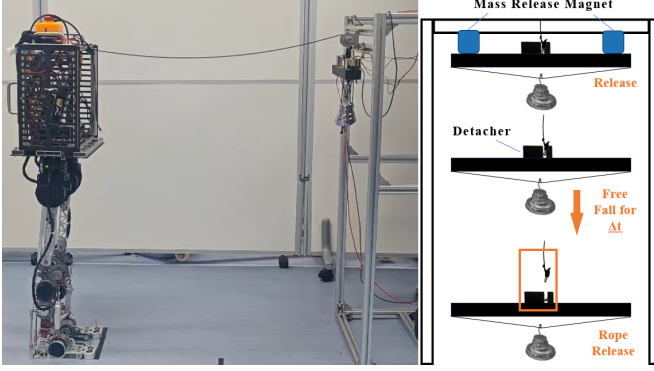


Fig. 6. Impact experiment configuration.

Based on this grouping setup, we designed and conducted two experiments. In the impact experiment, as shown in Fig. 6, we used the same impulse-generation apparatus as that described in [1] to enable standardized comparison. The impulse-generation apparatus first releases a suspended mass using an electromagnetic mass-release mechanism. As the mass falls, it pulls the robot through a rope, thereby applying an impulse to the robot. For each impulse level, five repeated trials were conducted, and the mean time required to recover static stability across the five trials was used as the evaluation metric. A trial was classified as a failure of static stability recovery if the robot either failed to recover stability within 10s after the impact or took a step to maintain balance.

Subsequently, we designed a walking experiment to investigate the respective effects of the active toe joint and the spring-tendon mechanism on gait performance and walking energy efficiency. In each trial, the robot was operated via a joystick to complete three 7-meters walking trials on a flat outdoor stone-paved surface at a preset speed of 0.7 m/s. The average power consumption across these repetitions was then calculated from the measured motor torque and angular velocity data. Additionally, the robot trajectory was recorded using a Real-Time Kinematic (RTK) GPS device mounted on top of the robot, enabling quantitative evaluation of the tracking error.

B. Impact Experiment

Fig. 7 illustrates representative results from the impact experiment. Across both BioLeg and Roboto, Group 1 consistently exhibited longer static-stability recovery times than Groups 2 and 3. Because the primary structural difference between Group 1 and the other two groups is the absence of an active toe joint, we infer that its poorer recovery performance is mainly attributable to the lack of active toe actuation. After the active toe joint was introduced, the recovery performance of Groups 2 and 3 became relatively similar, although Group 3 generally performed slightly better than Group 2. These results

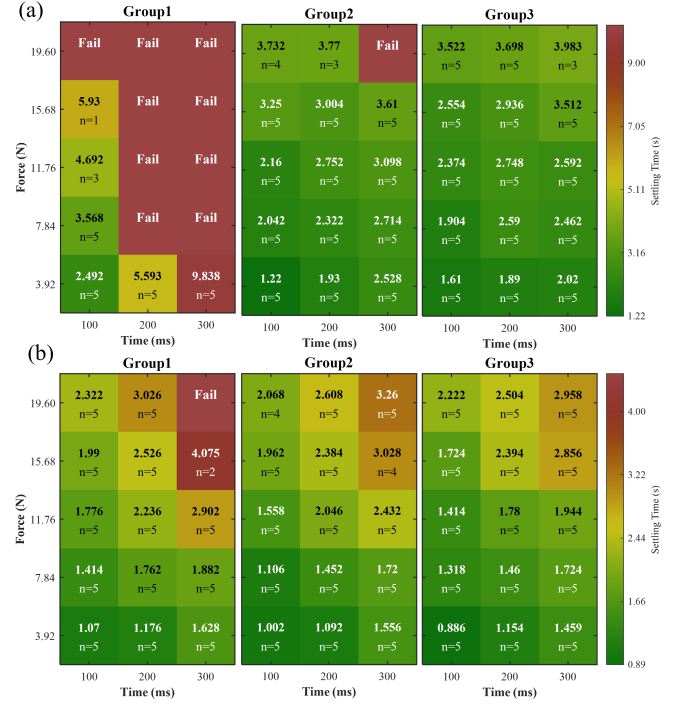


Fig. 7. Results of the impact experiments. n denotes the number of successful trials out of five repetitions. (a) Results for BioLeg. (b) Results for Roboto.

suggest that the spring-tendon mechanism can further improve static stability to some extent, while the presence of an active toe joint remains the dominant structural factor affecting static-stability recovery.

C. Walking Experiment

Fig. 8 presents the walking performance of the three experimental configurations on both robotic platforms. Using BioLeg as a representative example, it also shows the ankle and toe joint torque profiles over two gait cycles, together with the corresponding average mechanical power. These results are used to further investigate the underlying working mechanism of STERS and its effects on locomotion performance.

As shown in Fig. 8(a), Groups 2 and 3 exhibited better straight-line walking trajectories than Group 1 on both robotic platforms. The reduced lateral deviation is attributed primarily to improved heading preservation. By extending the effective support phase and regulating terminal-stance propulsion, the articulated toe produces smoother contact transitions, thereby suppressing the accumulation of yaw error and lateral drift. The STERS-enabled configuration (Group 3) exhibited the smallest terminal lateral deviation during the walking trial. This improvement may be attributed to the coupled elastic pathway, which smooths energy release during terminal stance and improves the consistency of propulsion between the two legs.

As shown in Fig. 8(b), Group 3 exhibited the lowest energy consumption on both BioLeg and Roboto, followed by Group 2, whereas Group 1, which was equipped with neither an active toe nor spring-tendon mechanism, consumed the most energy. Over the 7-meters walking trial, the complete STERS

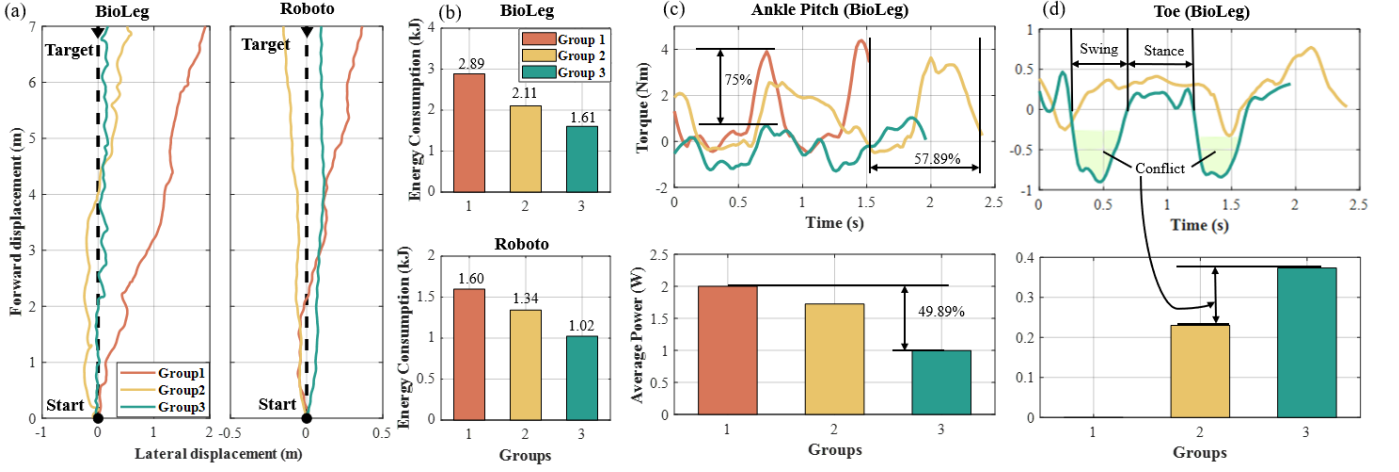


Fig. 8. Comparison of walking performance among the three groups. (a) Walking trajectories of BioLeg and Roboto. (b) Total energy consumption of the three groups during the 7-meters walking trials on BioLeg and Roboto. (c) Ankle pitch torque profiles over two gait cycles and the corresponding average power for BioLeg. (d) Toe joint torque profiles over two gait cycles and the corresponding average power for BioLeg.

configuration reduced the leg energy consumption of BioLeg and Roboto by 44.29% and 36.25%, respectively, relative to Group 1. Further analysis of the contribution of each component showed that introducing the active toe reduced energy consumption by 26.98% on BioLeg and 16.25% on Roboto. Building upon the active-toe configuration, the addition of the spring-tendon mechanism yielded further reductions of 23.70% and 23.88%, respectively.

To further investigate the mechanisms through which the active toe joint and the spring-tendon mechanism reduce energy consumption, we recorded the torque and mechanical power profiles of the ankle-pitch and toe joints. Using BioLeg as a representative example, the corresponding results are presented in Fig. 8(c) and (d).

First, in terms of gait timing, the introduction of the active toe joint did not produce a substantial reduction in the average mechanical power over a gait cycle. However, it increased the gait-cycle duration by 57.89%, thereby reducing the walking cadence. The lower cadence enabled smoother and more stable transitions between successive stance phases and have reduced the number of steps required to cover a fixed walking distance. Therefore, the energy-saving effect of the active toe joint appears to arise primarily from its regulation of the overall gait timing and propulsion pattern, rather than from a direct reduction in the instantaneous average power.

Second, the ankle-torque profiles indicate that the active toe joint mitigates abrupt changes in ankle torque at foot contact to some extent, thereby producing a smoother initial-contact transition. With the further addition of the spring-tendon mechanism to the active-toe configuration, the peak driving torque at the ankle was reduced by approximately 75% relative to Group 2. This result indicates that the coupled elastic pathway supports part of the ankle load during stance and reduces the instantaneous output required from the ankle actuator through elastic energy storage and release. In addition to reducing the mechanical power demand during high-load phases, the lower peak torque may also reduce motor losses associated with high current, thereby further improving the overall energy efficiency.

However, further analysis of Fig. 8(c) indicates that STERS may also introduce a local energetic penalty during specific phases of the gait cycle. By separating the toe-joint torque profiles of Groups 3 into stance and swing phases, we observe that, during stance, Group 3, which incorporates the complete STERS, requires less toe-joint torque than Group 2, which is equipped only with an active toe. This suggests that the spring-tendon mechanism assists the toe actuator by supporting part of its load during stance. During swing, however, Group 3 exhibits a pronounced increase in toe-joint torque, which is also reflected in its higher average mechanical power at the toe joint.

We hypothesize that this local increase in power consumption arises because the passive elastic torque generated by the spring-tendon mechanism opposes the active motor torque at the toe joint during swing. Consequently, the toe actuator must perform additional mechanical work to overcome the elastic torque and track the desired toe trajectory. Nevertheless, this local energetic penalty does not offset the overall energy-saving effect of STERS. During the ankle-lift phase, the toe-side moment arm is only approximately 37% of the ankle-side moment arm in kangaroos and 20% in BioLeg and Roboto. Moreover, the angular displacement of the ankle joint during walking is approximately two to three times that of the toe joint. Energy transfer through the spring-tendon mechanism is therefore dominated by the ankle side, and the additional local power consumed at the toe joint during swing remains relatively small compared with the overall reduction in locomotion energy consumption.

This study still has some limitations. Because the system employs fixed elastic stiffness and continuous ankle-toe coupling, conflicts between the passive elastic torque and the active joint-control torque cannot be completely avoided during swing. Future work will investigate elastic energy-transfer mechanisms with adjustable stiffness or switchable coupling to reduce undesired elastic torques and the associated energy losses. The adaptability and generalization capability of STERS will also be further evaluated across a wider range of walking speeds, terrain conditions, and dynamic locomotion

tasks.

VII. CONCLUSION

This paper presents STERS, an active-toe-coupled spring–tendon energy recirculation system, and integrates its dynamics model and multi-contact-stage gait guidance into a reinforcement learning framework. Unlike existing approaches that confine elastic elements to a single joint or treat the toe as an independent degree of freedom, STERS couples the ankle and active toe joints through a shared elastic pathway, enabling cross-joint energy storage, transfer, and release while maintaining stable autonomous walking. Ablation experiments on BioLeg and Roboto show that the complete STERS configuration achieves the lowest walking energy consumption and the smallest lateral deviation on both bipedal platforms. These results support the transferability of the STERS mechanical design and its associated control method across bipedal platforms with different mechanical structures and baseline control frameworks.

REFERENCES

- [1] B. v. Marum, A. Shrestha, H. Duan, P. Dugar, J. Dao, and A. Fern, “Revisiting reward design and evaluation for robust humanoid standing and walking,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, Abu Dhabi, UAE, 2024, pp. 11 256–11 263.
- [2] W. Zhu, L. Qiao, Y. Lu, X. Rong, Y. Fan, Y. Li, R. Song, and G. Zhang, “Design and control of halfMan: A human-sized bipedal robot with high payload,” *IEEE/ASME Trans. Mechatronics*, vol. 31, no. 1, pp. 278 – 289, 2026.
- [3] M. Mihalec and J. Yi, “Balance gait controller for a bipedal robotic walker with foot slip,” *IEEE/ASME Trans. Mechatronics*, vol. 28, no. 4, pp. 2012–2019, 2023.
- [4] G. Mesesan, R. Schuller, J. Engelsberger, C. Ott, and A. Albu-Schäffer, “Unified motion planner for walking, running, and jumping using the three-dimensional divergent component of motion,” *IEEE Trans. Robotics*, vol. 39, no. 6, pp. 4443–4463, 2023.
- [5] D. W. Haldane, M. M. Plecnik, J. K. Yim, and R. S. Fearing, “Robotic vertical jumping agility via series-elastic power modulation,” *Sci. Robot.*, vol. 1, no. 1, p. eaag2048, 2016.
- [6] D.-Y. Kim, S.-H. Yun, J.-K. Lee, J. Yoon, D. Nam, C.-Y. Maeng, and Y.-J. Kim, “Hyperleg: Biomechanics-inspired high-dof leg and toe mechanism for highly dynamic motions,” in *Proc. IEEE Int. Conf. Robot. Autom.*, Yokohama, Japan, 2024, pp. 2456–2462.
- [7] A. Badri-Spröwitz, A. A. Sarvestani, M. Sitti, and M. A. Daley, “Birdbot achieves energy-efficient gait with minimal control using avian-inspired leg clutching,” *Sci. Robot.*, vol. 7, no. 64, p. eabg4055, 2022.
- [8] A. S. Mastrogeorgiou, Y. S. Elbahrawy, A. Kecske méthy, and E. G. Papadopoulos, “Slope handling for quadruped robots using deep reinforcement learning and toe trajectory planning,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, Las Vegas, NV, 2020, pp. 3777–3782.
- [9] A. Mazumdar, S. J. Spencer, C. Hobart, J. Salton, M. Quigley, T. Wu, S. Bertrand, J. Pratt, and S. P. Buerger, “Parallel elastic elements improve energy efficiency on the STEPPR bipedal walking robot,” *IEEE/ASME Trans. Mechatronics*, vol. 22, no. 2, pp. 898–908, 2017.
- [10] D. Tlalolini, C. Chevallereau, and Y. Aoustin, “Human-like walking: Optimal motion of a bipedal robot with toe-rotation motion,” *IEEE/ASME Trans. Mechatronics*, vol. 16, no. 2, pp. 310–320, 2011.
- [11] K. Nishiwaki, S. Kagami, Y. Kuniyoshi, M. Inaba, and H. Inoue, “Toe joints that enhance bipedal and fullbody motion of humanoid robots,” in *Proc. IEEE Int. Conf. Robot. Autom.*, Philadelphia, PA, 2002, pp. 3105–3110.
- [12] E. Kouchak and A. Mokhtarian, “Controllability analysis of a standing bipedal robot with active toe-joints,” in *2018 6th RSI International Conference on Robotics and Mechatronics*, 2018, pp. 343–348.
- [13] Y. Liu, X. Zang, S. Heng, Z. Lin, and J. Zhao, “Human-like walking with heel off and toe support for biped robot,” *Applied Sciences*, vol. 7, p. 499, 2017.
- [14] E. P. Snelling, A. A. Biewener, Q. Hu, D. A. Taggart, A. Fuller, D. Mitchell, S. K. Maloney, and R. S. Seymour, “Scaling of the ankle extensor muscle-tendon units and the biomechanical implications for bipedal hopping locomotion in the post-pouch kangaroo *macropus fuliginosus*,” *J. Anatomy*, vol. 231, no. 6, pp. 921–930, 2017.
- [15] C. Stecco, M. Corradin, V. Macchi, A. Morra, and R. D. Caro, “Plantar fascia anatomy and its relationship with achilles tendon and paratenon,” *J. Anatomy*, vol. 223, no. 6, pp. 665–676, 2013.
- [16] A. Singh, J. Zwirner, F. Templer, D. C. Kieser, S. Klima, and N. Hammer, “On the morphological relations of the achilles tendon and plantar fascia via the calcaneus: a cadaveric study,” *Sci. Rep.*, vol. 11, no. 1, 2021, article 5985.
- [17] B. Kiss, E. C. Gonen, A. Mo, A. Badri-Spröwitz, A. Buchmann, and D. Renjewski, “Gastrocnemius and power amplifier soleus spring-tendons achieve fast human-like walking in a bipedal robot,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, Kyoto, Japan, 2022, pp. 5202–5209.
- [18] X. Liu, A. Rossi, and I. Poulakakis, “SPEAR: A monopodal robot with switchable parallel elastic actuation,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, 2015, pp. 5142–5147.
- [19] L. Yu, H. Zhao, L. Qiao, S. Qin, G. Jin, and Y. Chen, “A 2-DoF switchable variable stiffness actuator for legged walking robots,” *IEEE/ASME Trans. Mechatronics*, vol. 31, no. 3, pp. 2864 – 2875, 2026.
- [20] M. Xu, Z. Zhou, Z. Wang, L. Ruan, J. Mai, and Q. Wang, “Bioinspired cable-driven actuation system for wearable robotic devices: Design, control, and characterization,” *IEEE Trans. Robotics*, vol. 40, pp. 520–539, 2024.
- [21] X. Yin, J. Yan, S. Wen, J. Zhang, and M. Pang, “Origami-based decoupling clutch achieves energy-efficient legged robots,” *IEEE Robot. Automat. Lett.*, vol. 8, no. 12, pp. 8058–8065, 2023.
- [22] D. F. B. Haeufle, M. D. Taylor, S. Schmitt, and H. Geyer, “A clutched parallel elastic actuator concept: Towards energy efficient powered legs in prosthetics and robotics,” in *Proc. IEEE Int. Conf. Biomed. Robot. Biomechat.*, Rome, Italy, 2012, pp. 1614–1617.
- [23] W. Fan, Z. Dai, W. Li, and T. Liu, “Load-carrying assistance of articulated legged robots based on hydrostatic support,” *IEEE Robot. Automat. Lett.*, vol. 9, no. 10, pp. 8274–8281, 2024.
- [24] F. Bjelonic, J. Lee, P. Arm, D. Sako, D. Tateo, J. Peters, and M. Hutter, “Learning-based design and control for quadrupedal robots with parallel-elastic actuators,” *IEEE Robot. Automat. Lett.*, vol. 8, no. 3, pp. 1611–1618, 2023.
- [25] J. Ding, P. Posthoorn, V. Atanassov, F. Boekel, J. Kober, and C. D. Santina, “Quadrupedal locomotion with parallel compliance: E-go design, modeling, and control,” *IEEE/ASME Trans. Mechatronics*, vol. 29, no. 4, pp. 2839–2848, 2024.
- [26] J. Ding, V. Atanassov, E. Panichi, J. Kober, and C. Della Santina, “Robust quadrupedal jumping with impact-aware landing: Exploiting parallel elasticity,” *IEEE Trans. Robotics*, vol. 40, pp. 3212–3231, 2024.
- [27] Q. Liao, B. Zhang, X. Huang, X. Huang, Z. Li, and K. Sreenath, “Berkeley Humanoid: A research platform for learning-based control,” in *Proc. IEEE Int. Conf. Robot. Autom.*, Atlanta, GA, 2025, pp. 2897–2904.